AN AUTO-ASSOCIATED RING ANNIHILATES EVERY PROPER FINITELY GENERATED IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Auto-associated ring, defined by the existence of a nonzero element killed by a power of every element of the maximal ideal. defined in the same section, source lines 3650-3691. Supplies the element x and the property (P) to be proved.
- A criterion identifying the kernel of the map to the product of the localisations at the generators of an ideal. the author cites Algebra, Lemma algebra-lemma-when-injective-covering at proof lines 3703-3705. Converts the membership of x in that kernel into a nonzero annihilator of the ideal.

Source reference. Original lemma 05GN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7067–7067. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0565: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6698–6698. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *An auto-associated ring R has the following property: (P) Every proper finitely generated ideal $I \subset R$ has a nonzero annihilator.*

Proof. By assumption there exists a nonzero element $x \in R$ such that for every $f \in \mathfrak{m}$ we have $f^n x = 0$. Say $I = (f_1, \ldots, f_r)$. Then x is in the kernel of $R \to \bigoplus R_{f_i}$.

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Hence we see that there exists a nonzero $y \in R$ such that $f_i y = 0$ for all i, see Algebra, Lemma 0565. As $y \in \text{Ann}_R(I)$ we win. $\square$